REMI model

REMI model: Bottom-up emissions inventories for cities with limited data (formerly remIAG)

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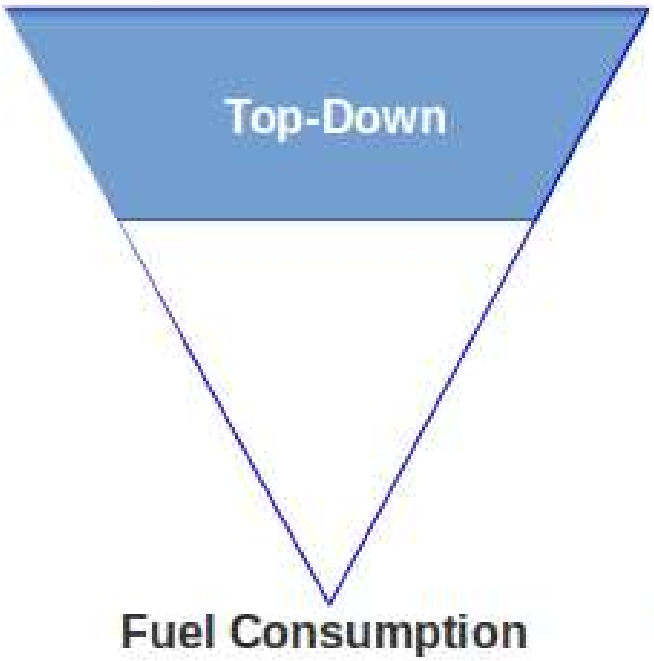
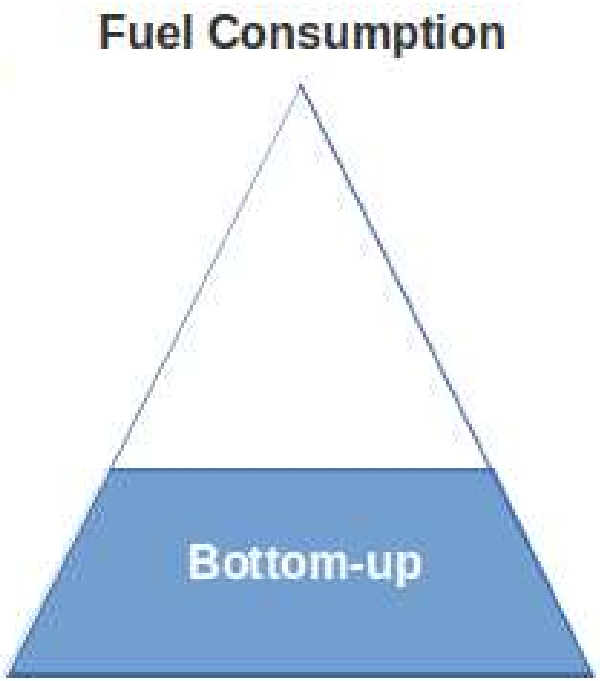
Emissions Inventories

- Emission Inventories are often considered to be the scapegoat for air quality assessment.
- Mismatch between modeled and observed concentrations of air pollutants.
- More detailed inventories are therefore required.

Two types of inventory approaches





- Based mainly on national/regional statistics

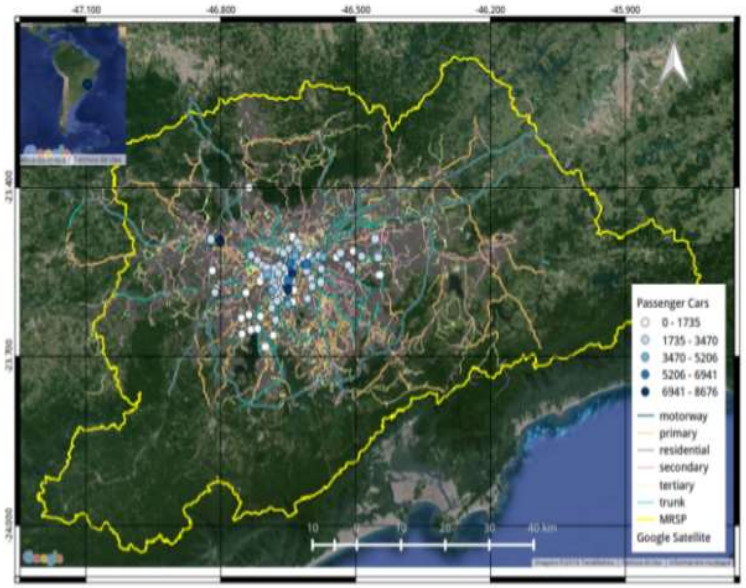
Based on local traffic counts, speed recordings, etc

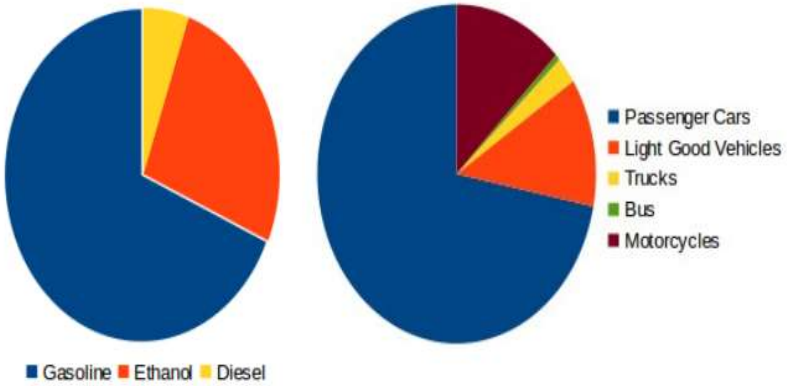
Study Area: Metropolitan Area of São Paulo

- Population: 20M
- Registered vehicles: 7M

Traffic information

- Traffic counts
- Speed recordings
- Fleet statistics
- Traffic model





R Emissions Inventory Model (REMI)

REMI model characteristics

Pollutants

- CO, NO_X, HC, $PM_{2.5}$, NO, NO_2, NH_3, CO_2, SO_2, N_2O, FC, NMHC (and 15 species)

Activity

- Interpolates traffic counts.
- Reads traffic model outputs.
- Fleet data is organized according to EU fleet format.

Emission factors (EF)

- Currently uses Brazilian EF.
- Scaled using COPERT.

Outputs

- Graphics and tables

REMI model summary

- REMI is written in R.
- There is a main routine that calls in sub-routines.
- Currently all user inputs are hardcoded (e.g. no user interface).
- Once you understand the code, it is relatively straightforward.

input data

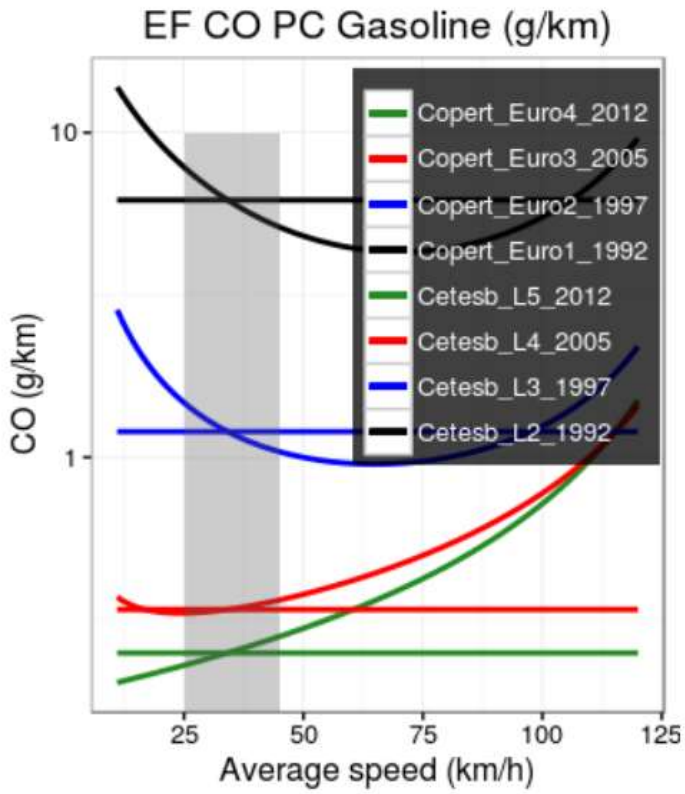
REMI Model: Input data

- **Road Links**
- **Open Street Maps**
- **Traffic flows**
- **24 hour flow variation profiles over 7 days**
- **Average speed**
- **Emission factors** (i.e. kilometers driven)
- **Fleet composition** (Vehicle type, size, weight, fuel, euro equivalence, age)

some functions

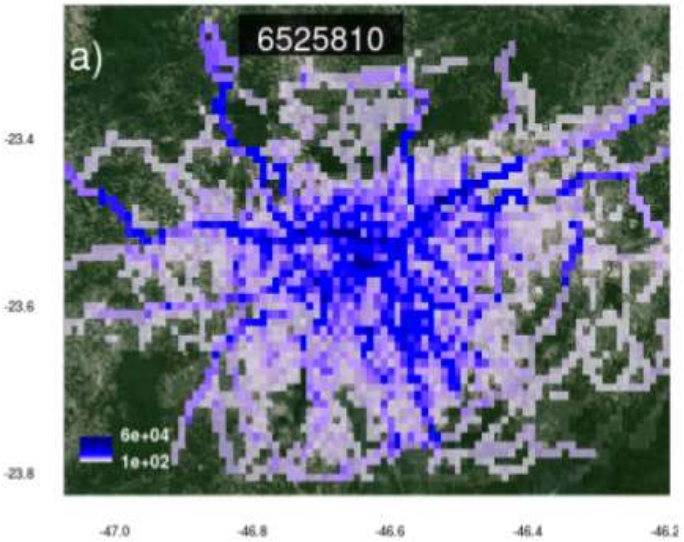
REMI functions

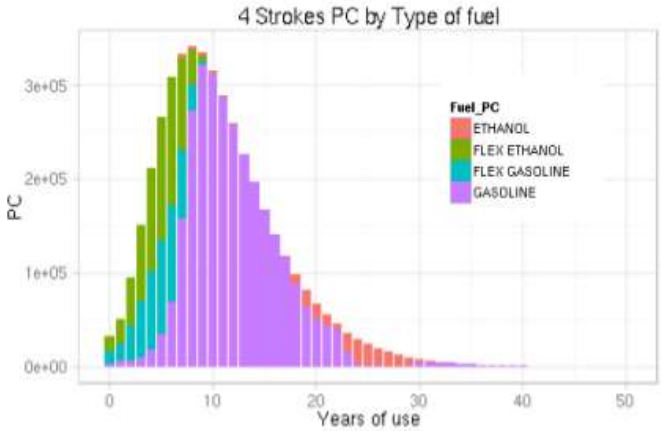
- Traffic interpolated to project flows module
- Fleet distribution module based on age profiles
- Emissions factor interpolation module
- Tabulated/graphical outputs
- Spatial outputs module

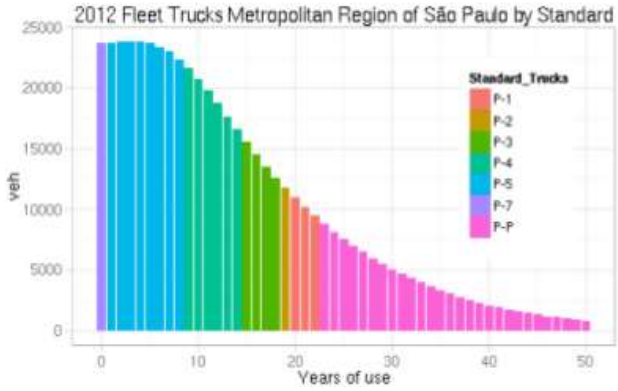


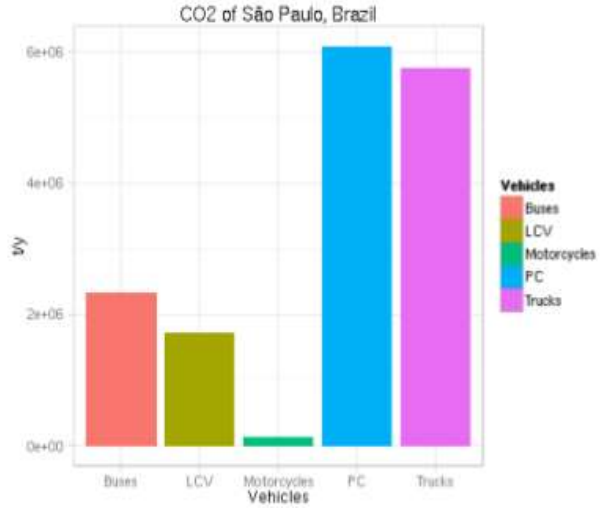
output examples

Graphical outputs

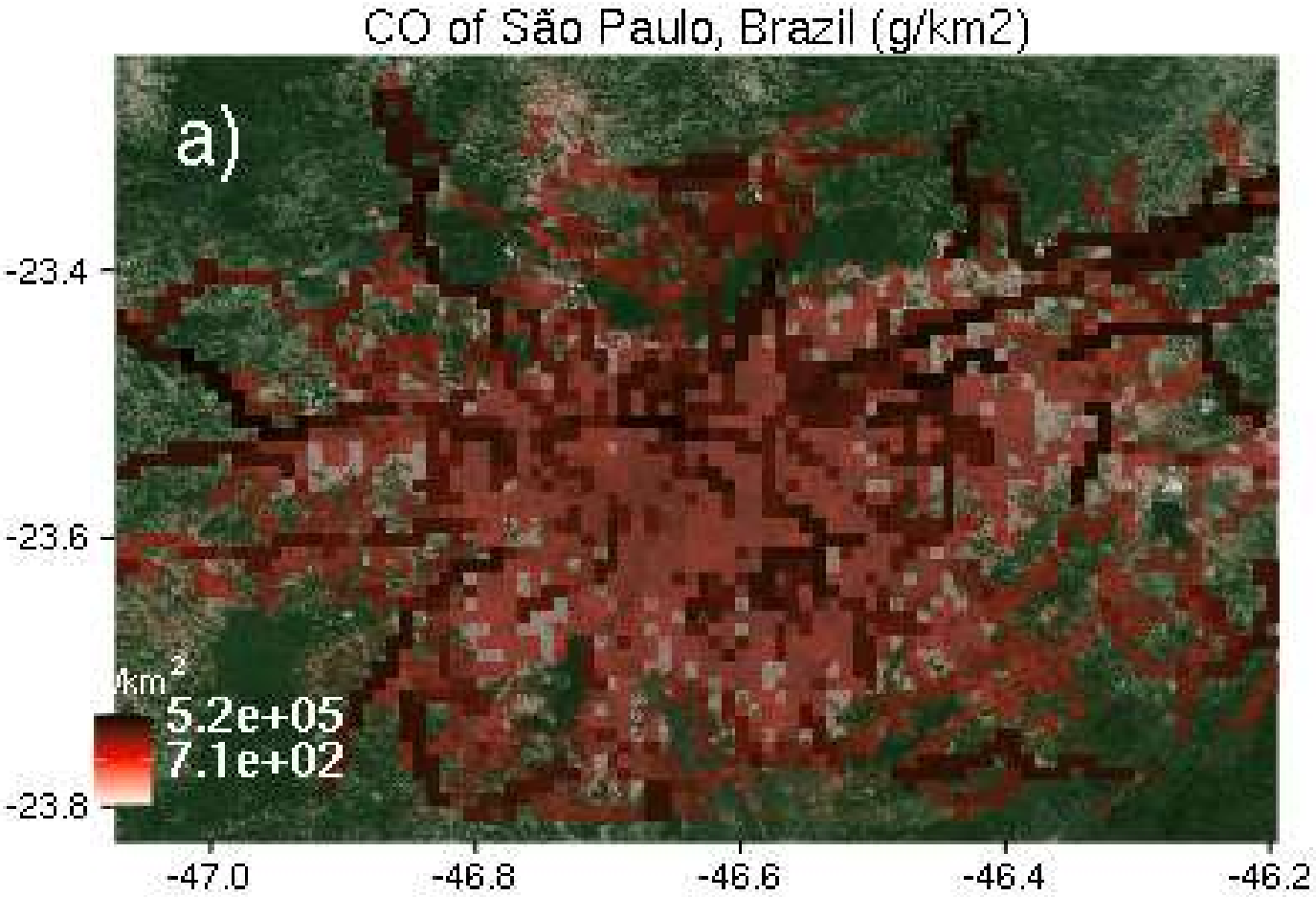




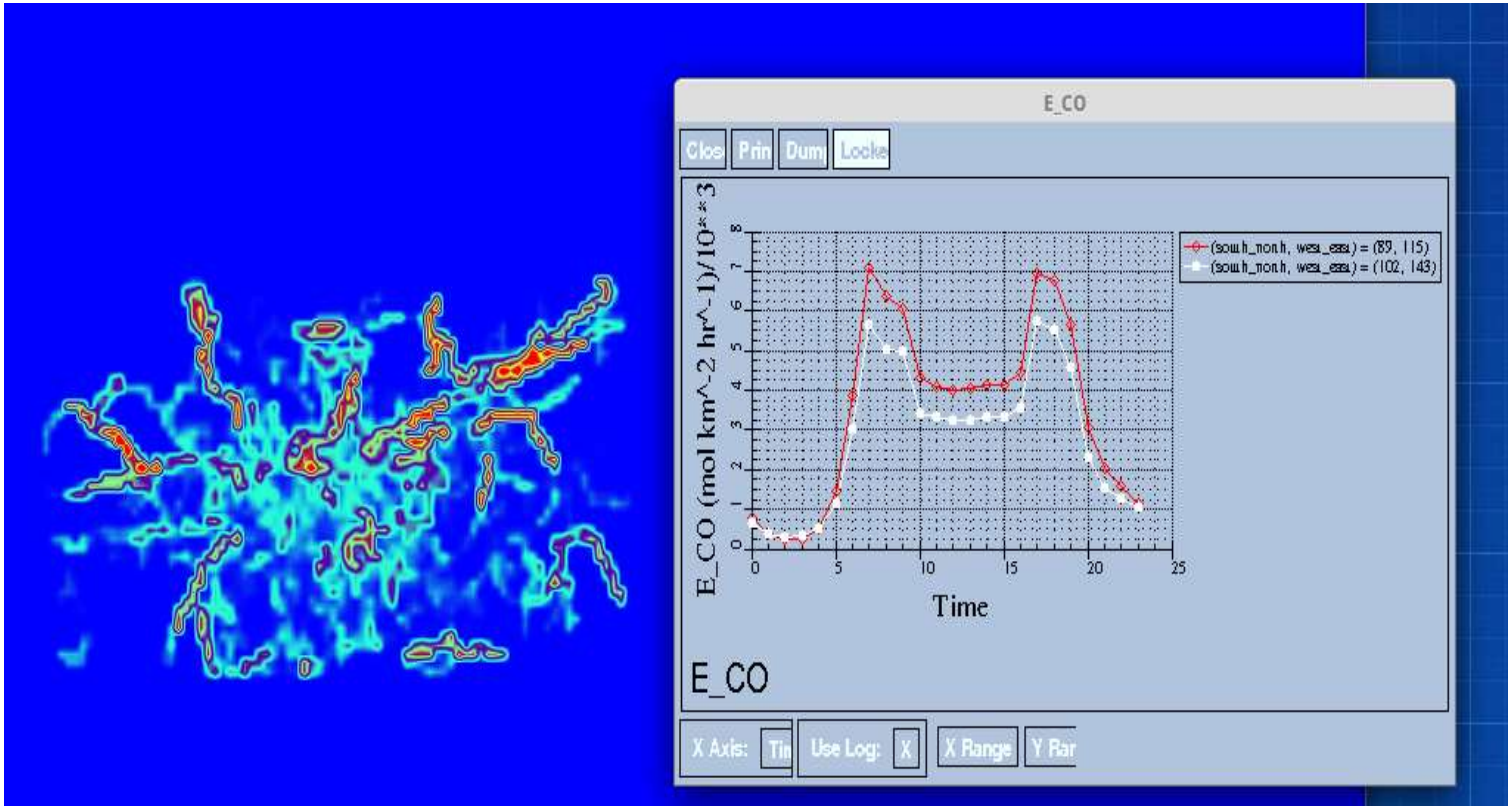




CO emissions 1 km^2



WRF-Chem Inputs



applications

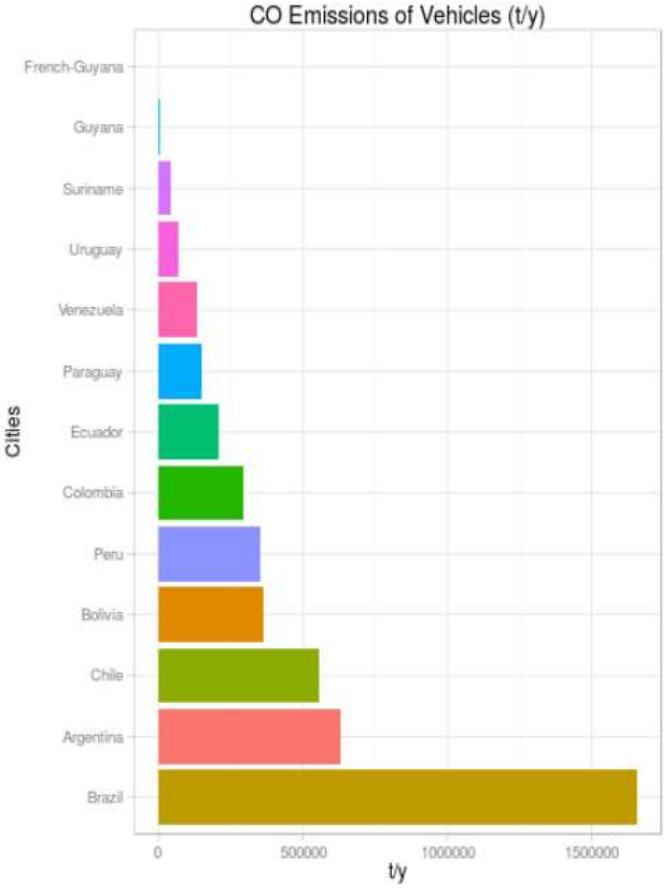
Comparison with fuel sales at Metropolitan Region of São Paulo

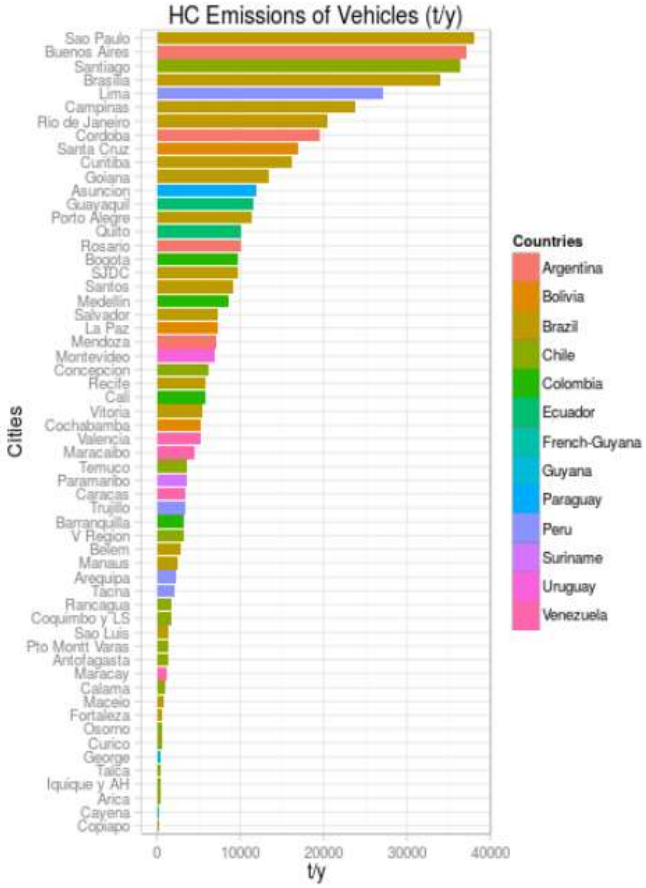
- Estimated and observed fuel consumption in MASP m^3

Traffic	Gasoline	Ethanol	Diesel
Consumption	4,763,370	2,192,740	3,334,037
Traffic count	2,943,859	1,003,151	2,504,437
%	61.80	45.75	75.12
Traffic model	4,119,984	1,361,545	4,423,238
%	86.49	62.09	132.67

- (Ibarra & Ynoue, 2016)

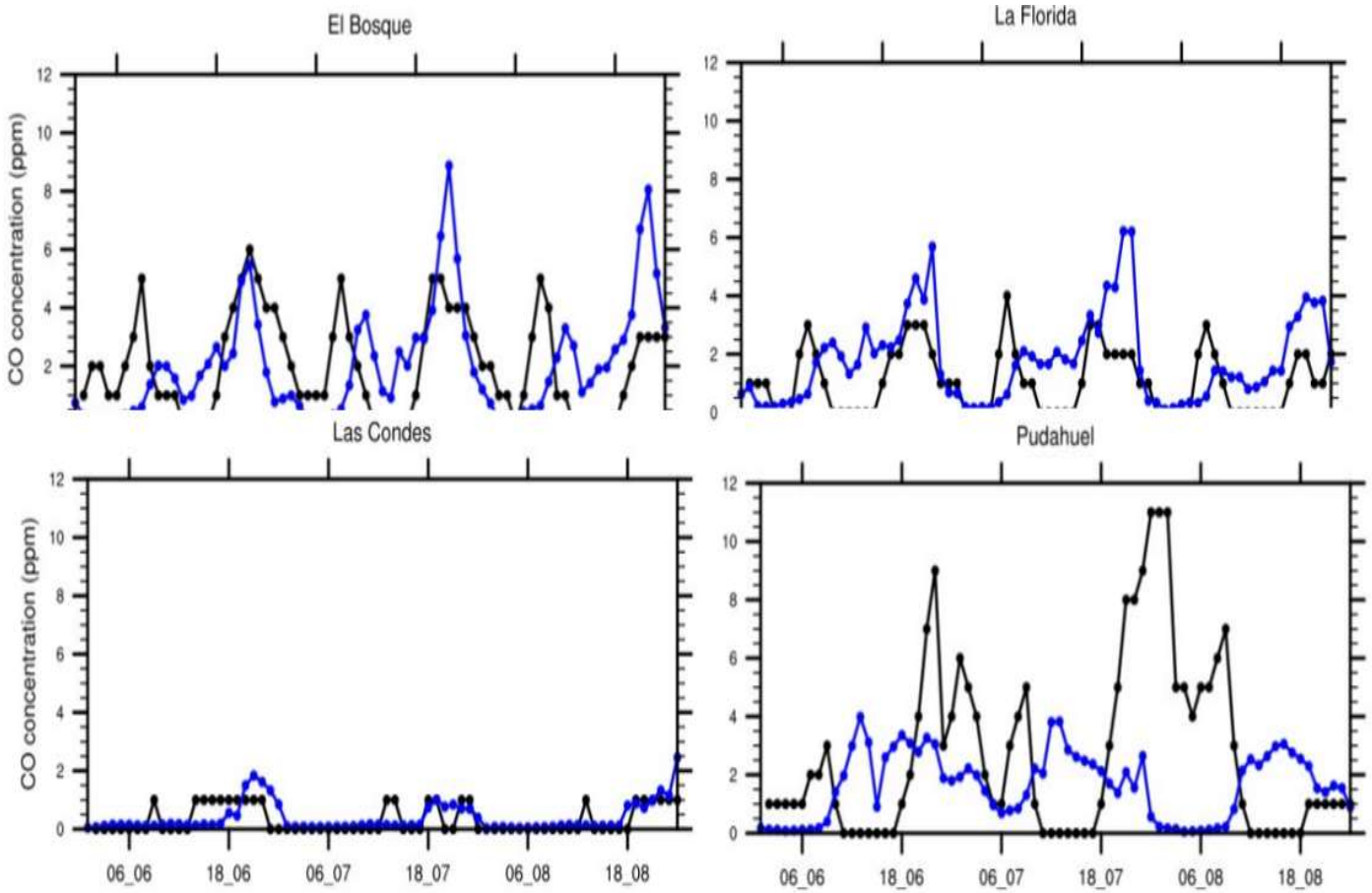
Emission estimates for 58 cities in South America





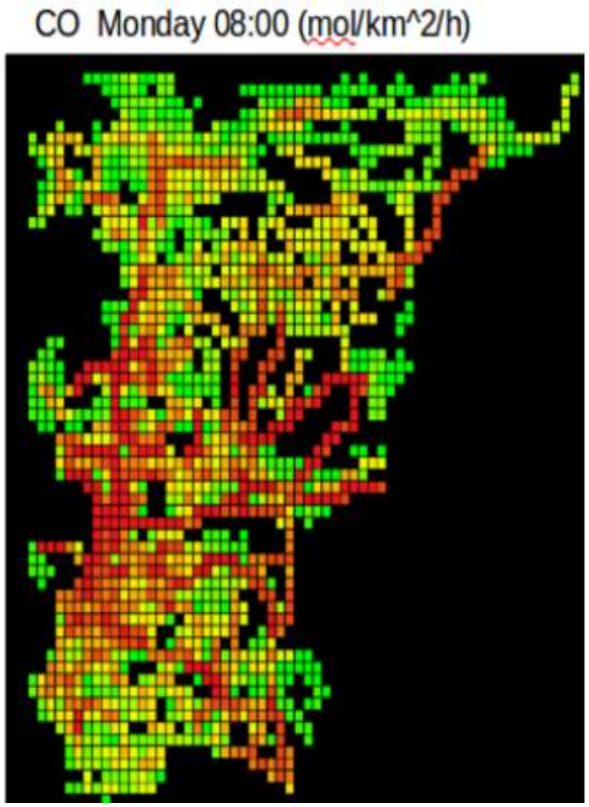
- (Ibarra et al, 2015a)

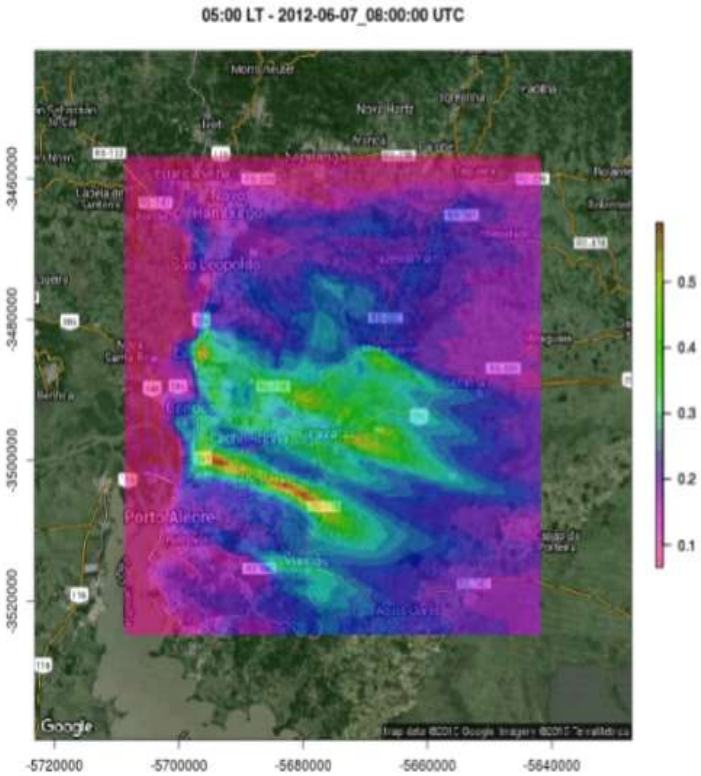
WRF-Chem simulation of CO in Santiago, Chile



- (Ibarra et al, 2015a)

WRF-Chem simulation Porto Alegre, Brazil





- (Ibarra et al, 2015b)

REMI performance

- REMI runs on any operational system
- Time to calculate emissions (including gridding and all plots):

REMI	Metropolitan Area of São Paulo (MASP)	Time
Count Interpolation	105468 links 27 pollutants 24 hours 7 days of the week	4 hours
Traffic model	34733 links 27 pollutants 24 hours 7 days of the week	1 hour

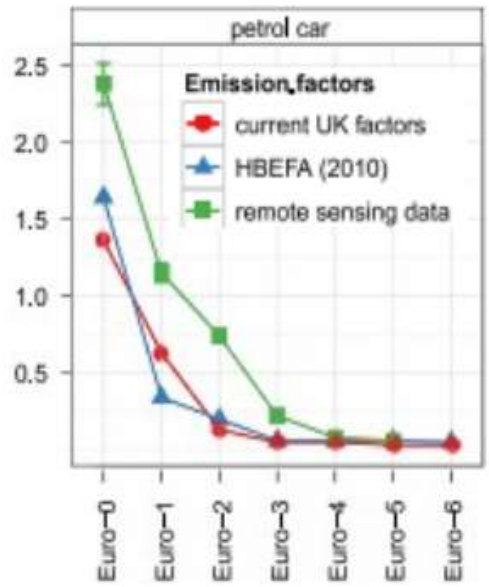
Limitations of REMI

- Currently it only works for Brazil and 2012 base-year
- Traffic interpolation is sensitive to road type and relative location to city centre.
- It is limited to non-motorway regions
- There is no user interface

next steps

Next steps

- Refine fleet composition.
- Fleet forecasting for future years (e.g. 2020).
- Scenario module.
- Integrate scaling based on real driving emissions (RDE) (Carslaw, 2011)
- Include emission factors, such as HBEFA
- Load different interpolations: krigging, spatial autocorrelation, etc.
- Include different type of pollution sources
- Global vehicular bottom-up emissions inventory.



Acknowledgment

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- Kevin Turpin, Anna Savage and Tim Barlow (TRL)

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thanks